Refining the Computational Analysis of Interactive Fiction: A Hybrid Framework using Bag-of-Paths Formalism and LLMs

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Introduction

Interactive Fictions and Gamebooks

An **interactive fiction** is a form of narrative in which the reader's or player's choices influence the sequence of events, the accessible content, or the story's outcome.

A **gamebook** is a book-based form of interactive fiction, usually divided into numbered sections connected by choices, rules, and sometimes game mechanics such as combat, inventory, or chance.

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Suddenly, the large rock you are hiding behind is rolled aside and you are faced by two snarling Giaks intent on your death. The cave mouth is a narrow entrance and you can only fight the Giaks one at a time.

Giak 1: COMBAT SKILL 13   ENDURANCE 10

Giak 2: COMBAT SKILL 12   ENDURANCE 10

If you win, you may explore the cave further by **turning to 33**.

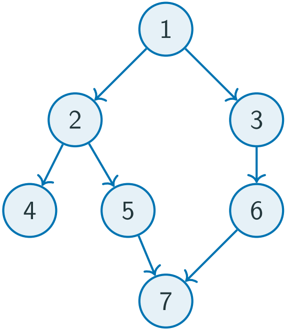
Or you may leave and descend the hill. **Turn to 248**.

Flight from the Dark by Joe Dever and Gary Chalk www.projectaon.org

The classical representation : a graph

Interactive fiction is traditionally modeled as a **directed graph** :

- **Nodes** represent passages or narrative sections ;
- **Edges** represent the choices available to the player.



Main limitation

This representation makes it possible to study the **structure** or the **topology** of the work, but the text itself, therefore the **narrative meaning**, is no longer taken into account.

A hybrid solution : graphs and LLMs

Large Language Models (LLM) make it possible to reintroduce the semantic content discarded by graph modeling.

Our approach : both methods

- The **graph** models the **structure** of the work
- The **LLM** interprets the **narrative content**

Gamebook text

↓

LLM annotation
Interpret the transitions offered to the player

↓

Graph modeling
Model the structure and produce representative stories

↓

LLM annotation
Analyse each resulting story as a whole

Corpus : Lone Wolf and Project Aon

The studied corpus is the **Lone Wolf series** created by Joe Dever.

The **project Aon** (www.projectaon.org) provides authorised digital editions of the books. It is already structured with html.

There are **several mechanics** in these gamebooks :

- Branching choices and forced transitions
- Random events
- Player profile with stats and special powers
- Combat with possible evasion
- Endurance (HP) management
- Items and inventory management
- Puzzles and hidden links

Scope of this presentation

A complete representation of every mechanic would make the model highly complex (it requires persistent states). We use **explicit simplifications** and focus exclusively on **Book 1 : Flight from the Dark**.